

## CE 310 — Week 15 Self-Check

**Week:** 15 | **Topic:** Capstone Readiness **10 questions** · **OPTIONAL AND UN-GRADED**

**This one does not count toward your grade.** The 13 graded weekly quizzes are in Weeks 1–7 and 9–14. Week 15 is exam week — you have the Final Written Methods Exam on Monday and the Python Practicum on Wednesday — so this is a self-check, not another graded deliverable.

Use it to find your weak spots before those two exams and before the capstone analysis. It covers regression, hypothesis testing, sensitivity, and Monte Carlo from Weeks 9–13. Answers are released immediately; take it as many times as you find useful.

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**Q1.** A multiple regression model gives  $\text{Cooling\_kWh} \sim 0.82 \cdot \text{Outdoor\_Temp\_F} + 3.45 \cdot \text{BH}$ . The BH coefficient is +3.45. What does this mean?

- A) For every 1°F increase in outdoor temperature, cooling increases by 3.45 kWh
- B) The model predicts 3.45 kWh of cooling when both Outdoor\_Temp\_F and BH equal zero
- C) Removing business hours reduces annual cooling by 3.45 kWh
- D) Business hours add 3.45 kWh of cooling at the same outdoor temperature

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**Q2.** [True/False]  $R^2$  can only rise when a predictor is added to a regression, so on its own it cannot tell you whether the addition was worthwhile.

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**Q3.** A multiple regression model for a small hotel returns a BH coefficient of -3.27. The hotel's standard business hours are M-F 9 AM–5 PM. Which interpretation is most physically plausible?

- A) The model is incorrect and should be re-estimated
- B) Hotels have no cooling demand during business hours
- C) At the same outdoor temperature, daytime M-F hours use less cooling than evenings and weekends

- D) The negative sign means the BH variable was entered in the wrong direction in the formula
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**Q4.** A two-sample t-test compares mean cooling during business hours (mean = 34.0 kWh/hr) vs. non-business hours (mean = 29.8 kWh/hr) and gives  $p = 8.45 \times 10^{-42}$ . The regression BH coefficient is  $-3.27$ . Why do the t-test and regression disagree on the direction of the BH effect?

- A) The t-test uses a different dataset than the regression
  - B) The regression formula should use OLS; the t-test uses a different estimator
  - C) The p-value is too small for the t-test to be valid
  - D) The t-test does not control for outdoor temperature; the regression does
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**Q5.** [Matching] Match each method to the question it answers.

Choices:

- (A) Two-sample t-test
- (B) One-way ANOVA
- (C) Multiple regression
- (D) Monte Carlo simulation
- (E) Bootstrap

Matches:

1. How does the outcome change with several predictors at once, holding the others fixed?
  2. What interval should I put around this statistic, given only my data?
  3. Do these two groups differ by more than chance would produce?
  4. What is the distribution of the outcome when the inputs are uncertain?
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**Q6.** You run an OAT (one-at-a-time) sensitivity analysis on a regression model and find that a  $\pm 1$  standard deviation change in outdoor temperature changes annual cost by  $\pm \$8,200$ , while a  $\pm 1$  standard deviation change in the electricity rate changes annual cost by  $\pm \$7,100$ . What does this suggest?

- A) Both inputs contribute approximately equally to cost uncertainty
- B) Electricity rate is the dominant driver of cost uncertainty

- C) Temperature is the dominant driver of cost uncertainty
  - D) Neither input drives cost uncertainty — the model intercept dominates
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**Q7.** [Select all that apply] Adding a predictor raises  $R^2$  from 0.811 to 0.813 while adjusted  $R^2$  falls from 0.811 to 0.810. Which conclusions are supported?

- (A) The new predictor explains almost nothing once its cost in complexity is counted
  - (B) The rise in  $R^2$  is uninformative here, because  $R^2$  cannot fall when a predictor is added
  - (C) The model is now certainly overfit and unusable
  - (D) Dropping the new predictor is the reasonable default
  - (E) Adjusted  $R^2$  is not valid for models with more than two predictors
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**Q8.** [Ordering] Put the stages of a capstone analysis in the order you would carry them out.

- (A) Explore the data — distributions, a scatter plot, a comparison between the two groups
  - (B) Fit a regression and interpret the coefficients in the units of the problem
  - (C) Test whether the group difference you observed is larger than chance would produce
  - (D) Quantify what remains uncertain, with a bootstrap interval or a Monte Carlo range
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**Q9.** Two capstone teams both use temperature as a predictor variable, but analyze completely different topics (one studies cooling energy, another studies traffic volume). Both relationships are statistically significant, but the two teams get very different  $R^2$  values. What best explains this?

- A) Different domains simply have different amounts of unexplained variance
  - B) One of the two teams made a calculation error
  - C)  $R^2$  should always come out identical if the regression is done correctly
  - D) Random noise in data collection accounts for all of the difference
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**Q10.** After fitting a regression model, you plot residuals (actual – predicted) vs. fitted values and see a fan shape (residuals spread wider at high fitted values). What does this indicate?

- A) The model has high  $R^2$  and is a good fit
  - B) The model may have heteroskedasticity — variance of errors increases with the fitted value
  - C) The model has too many predictors
  - D) The model is correctly specified and the fan shape is expected
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